Sales Performance Analysis

1. Objective

The objective of this project is to analyze Superstore sales data, identify sales trends and high-performing products, evaluate regional performance, and generate actionable business insights using Python and Tableau.

2. Dataset Information

Dataset Name: Superstore Sales Dataset

Source: Kaggle

Dataset Description:

- The dataset contains sales transaction records from a retail superstore.
- It includes information about customer orders, products, sales, shipping details, and geographical regions.
- The dataset is used to analyze sales performance, identify trends, compare regional sales, and determine top-performing products.

Key Columns Used:

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Region
- State
- Category
- Sub-Category
- Product Name
- Sales

Total Records: *9800*

Total Columns: 18

3. Data Cleaning

```python
import pandas as pd
df = pd.read_csv("superstoresales.csv", encoding="latin1")
df.head()
```

[3] 0.8s Python

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798	Furniture	
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454	Furniture	
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240	Office Supplies	
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-TA-10000577	Furniture	
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-ST-10000760	Office Supplies	

Figure 1:Dataset Preview

Observation

- The first five records of the dataset were displayed successfully.
- The dataset contains customer, order, product, shipping, and sales information.
- The data appears to be properly structured and ready for analysis.

```
df.info()
```

```
[4] 0.0s
<class 'pandas.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row_ID         9800 non-null   int64
 1   Order_ID       9800 non-null   str
 2   Order_Date     9800 non-null   str
 3   Ship_Date      9800 non-null   str
 4   Ship_Mode      9800 non-null   str
 5   Customer_ID    9800 non-null   str
 6   Customer_Name  9800 non-null   str
 7   Segment        9800 non-null   str
 8   Country        9800 non-null   str
 9   City           9800 non-null   str
 10  State          9800 non-null   str
 11  Postal_Code    9789 non-null   float64
 12  Region         9800 non-null   str
 13  Product_ID     9800 non-null   str
 14  Category       9800 non-null   str
 15  Sub_Category   9800 non-null   str
 16  Product_Name   9800 non-null   str
 17  Sales          9800 non-null   float64
dtypes: float64(2), int64(1), str(15)
memory usage: 1.3 MB
```

Figure 2:Dataset Information

Observation

- The dataset contains **9,800 rows** and **18 columns**.
- Most columns are of **string (object)** data type, while **Sales** and **Postal_Code** are numeric.
- Only the **Postal_Code** column contains missing values (11 records), whereas all other columns are complete.

```
[4]  df.isnull().sum()

...  Row_ID            0
     Order_ID          0
     Order_Date        0
     Ship_Date         0
     Ship_Mode         0
     Customer_ID       0
     Customer_Name     0
     Segment           0
     Country           0
     City              0
     State             0
     Postal_Code      11
     Region            0
     Product_ID        0
     Category          0
     Sub_Category      0
     Product_Name      0
     Sales             0
     dtype: int64
```

Figure 3:Missing Values

Observation

- Missing value analysis shows that only the **Postal_Code** column has **11 missing values**.
- All remaining columns have **no missing values**, indicating good overall data quality.

4. Summary Statistics

```
[8]  df.mean(numeric_only=True)

...  Row_ID           4900.500000
     Postal_Code     55273.322403
     Sales             230.769059
     dtype: float64
```

Figure 4:Mean Output

```
[9]  df.median(numeric_only=True)

...  Row_ID          4900.50
     Postal_Code    58103.00
     Sales             54.49
     dtype: float64
```

Figure 5:Median Output

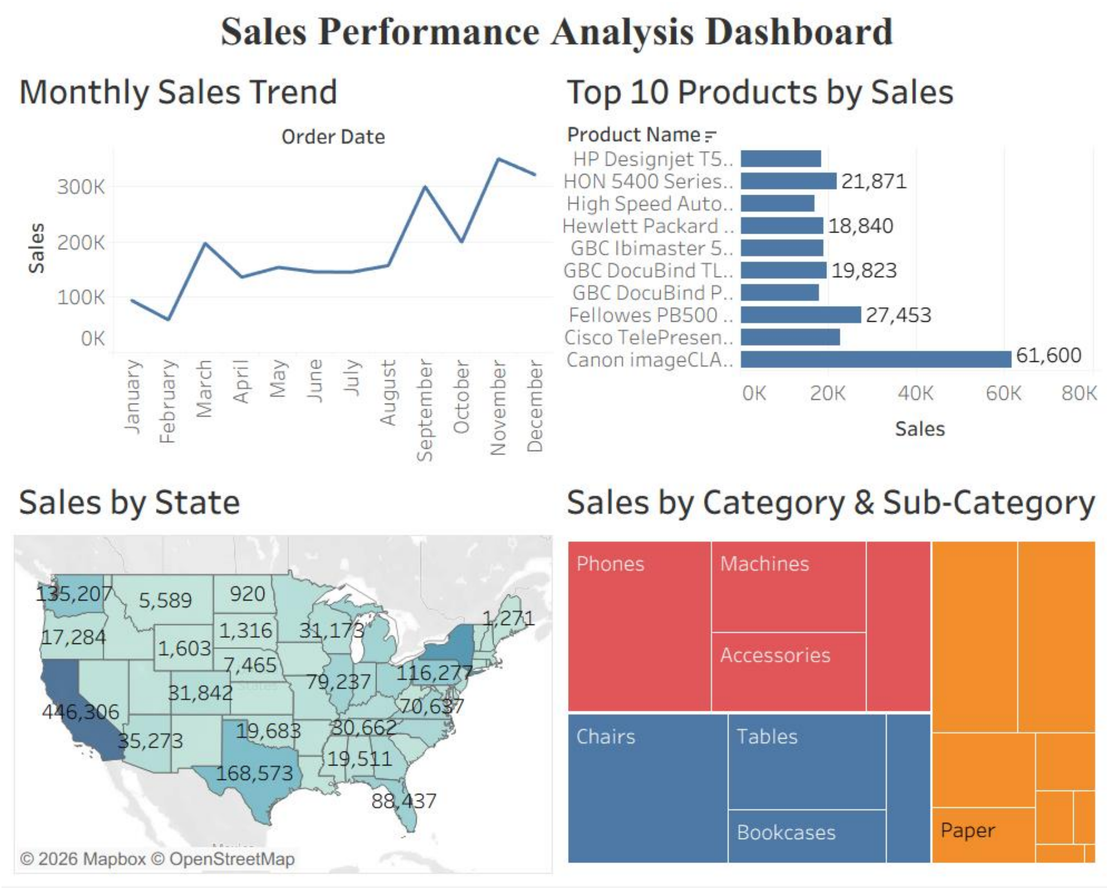
```
[10]  df.mode().iloc[0]

...   Row_ID                             1
      Order_ID              CA-2018-100111
      Order_Date                  5/9/2017
      Ship_Date                 26/09/2018
      Ship_Mode             Standard Class
      Customer_ID                 WB-21850
      Customer_Name          William Brown
      Segment                     Consumer
      Country                United States
      City                   New York City
      State                     California
      Postal_Code                  10035.0
      Region                          West
      Product_ID           OFF-PA-10001970
      Category             Office Supplies
      Sub_Category                 Binders
      Product_Name         Staple envelope
      Sales                          12.96
      Name: 0, dtype: object
```

Figure 6:Mode Output

5. KPIs

```
[13]  df["Sales"].sum()

...   np.float64(2261536.7827)

[14]  df["Sales"].mean()

...   np.float64(230.76905945918367)

[15]  df["Sales"].max()

...   np.float64(22638.48)

[16]  df["Sales"].min()

...   np.float64(0.444)

[6]   df["Order_ID"].nunique()
      ✓ 0.0s

...   4922
```

6. Visulaizations



1. **Monthly Sales Trend**
    1. Shows monthly variation in sales.
    2. Helps identify seasonality and peak sales months.
2. **Top 10 Products by Sales**
    1. Displays the products generating the highest revenue.
    2. Identifies best-selling products.
3. **Sales by State**
    1. Visualizes regional sales distribution across states.
    2. Highlights high-performing and low-performing regions.
4. **Sales by Category & Sub-Category**
    1. Shows contribution of different product categories.
    2. Helps compare category-wise sales performance.

7. Executive Summary

The sales analysis of the Superstore dataset reveals that sales vary significantly across products, categories, and regions. Certain products contribute a major share of revenue, while sales are concentrated in a few states. Monthly sales trends indicate seasonal fluctuations, with higher sales observed during the end of the year. The dashboard provides valuable insights that can help improve product planning, inventory management, and regional sales strategies.

8. Five Tactical Improvements

- Increase inventory for top-selling products to avoid stock shortages.
- Focus marketing campaigns on low-performing regions to improve sales.
- Promote under-performing product categories through discounts and bundle offers.
- Plan inventory based on seasonal sales trends to meet customer demand.
- Use sales insights to recommend popular products and improve customer purchasing experience.